

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

Smart Urban Mobility and Traffic Intelligence Dashboard

Batch No: 3

Prepared by:

- 1) Lahari Guggilam
- 2) Debarati Saha
- 3) Kajal G
- 4) Mareedu BhavyaSree
- 5) Preeti Sharma

Guide: Kalaiyarasi N

Tool Used: Microsoft Power BI Desktop

Period of Internship: August 2026 – October 2026

Milestone 1 — Data Modeling, DAX Measures & Dashboard Design

➤ Objective

The objective of Milestone 1 is to build the foundational structure of the Smart Urban Mobility and Traffic Intelligence Dashboard in Power BI. This includes preparing the dataset, designing a proper data model, writing DAX measures, and building an interactive, presentable dashboard page that satisfies the following requirements set by the project guide:

- A suitable background and clear heading for the dashboard
- Relevant slicers for interactive filtering
- Four KPI cards summarizing the dataset
- Pie/donut chart(s)
- A detailed table
- Other relevant visuals (bar chart, line chart)
- Relationship creation between tables
- DAX functions/calculations

1. Dataset Overview

The dataset used is Smart_City_Mobility_Sample_500.xlsx — a 500-row random sample drawn from a larger 200,000-row Smart City Mobility dataset. Each row represents a single recorded trip.

1.1 Key Columns

Column	Description
Route ID	Unique identifier for a travel route
Location	City and region combined (e.g., 'Tokyo - West')
Vehicle ID / Vehicle Type	Identifies the vehicle and its category (EV, Bus, Car, Taxi, Motorcycle)
Transport Mode	Public, Shared, or Private
Average Speed (KMH)	Recorded average speed for the trip
Passenger Count	Number of passengers on the trip
Temperature (C) / Weather / Rainfall (MM)	Environmental conditions during the trip
Date	Date the trip was recorded
Population* / Population Density*	Proxy metrics (Urban Growth Score / Traffic Density) substituted for true population data, which was not present in the original source
Vehicle Count*	Traffic-volume proxy — total trips recorded for that Route ID across the full dataset

2. Data Cleaning & Preparation

- Loaded the Excel dataset into Power BI Desktop.
- Split the single flat sheet into a star schema: a central fact table (Fact_mobility) and lookup/dimension tables (Dim_location, Dim_route).
- Removed duplicate values from Dim_location and Dim_route in Power Query so each dimension table holds one unique row per value, as required for valid one-to-many relationships.
- Verified column data types (Date, Text, Decimal/Whole Number) were correctly assigned.

3. Data Modeling & Relationships

A star schema was built with Fact_mobility at the center, connected to three dimension tables:

From Table	To Table	Key Column	Cardinality	Cross-filter
Dim_location	Fact_mobility	Location	One to many (1:*)	Single
Dim_route	Fact_mobility	Route ID	One to many (1:*)	Single
DateTable	Fact_mobility	Date	One to many (1:*)	Single

An additional DateTable was created using DAX to support time-based analysis, with the following calculated columns:

```
DateTable = CALENDAR(MIN(Fact_mobility[Date]), MAX(Fact_mobility[Date]))
Year = YEAR(DateTable[Date])
MonthNum = MONTH(DateTable[Date])
MonthName = FORMAT(DateTable[Date], "MMM")
```

The MonthName column was sorted by MonthNum (via Column Tools → Sort by Column) so that the Line Chart and Month slicer display months in calendar order rather than alphabetically.

4. DAX Measures

Measure	DAX Formula	Purpose
Total Trips	COUNTROWS(Fact_mobility)	Counts total recorded trips
Avg Speed	AVERAGE(Fact_mobility[Average Speed (KMH)])	Average trip speed
Total Passengers	SUM(Fact_mobility[Passenger Count])	Sum of all passengers carried
Total Vehicle Count	SUM(Fact_mobility[Vehicle Count*])	Traffic-volume proxy total
Avg Temperature	AVERAGE(Fact_mobility[Temperature (C)])	Average recorded temperature
Avg Rainfall	AVERAGE(Fact_mobility[Rainfall (MM)])	Average recorded rainfall

Distinct Routes	DISTINCTCOUNT(Fact_mobility[Route ID])	Number of unique routes
Distinct Locations	DISTINCTCOUNT(Fact_mobility[Location])	Number of unique locations

5. Dashboard Design

Requirement	What Was Done	Status
Suitable background	Applied a Power BI theme (navy/teal) via View → Themes for a consistent, professional look	Done
Clear heading	Text box: “Smart Urban Mobility and Traffic Intelligence Dashboard”, bold, 28pt	Done
Relevant slicers	Vehicle Type, Weather, and Month (sorted in calendar order)	Done
4 KPI cards	Total Trips, Avg Speed, Total Passengers, Total Vehicle Count	Done
Pie/Donut chart	Donut chart: Total Trips by Transport Mode (Public/Shared/Private)	Done
Table	Route-level detail table: Route ID, Location, Vehicle Type, Avg Speed, Passenger Count	Done
Other relevant visuals	Bar chart: Total Trips by Vehicle Type; Line chart: Total Trips by Month	Done
Relationships between tables	Three 1:* relationships connecting dimension tables to the fact table	Done
DAX functions/calculations	8 DAX measures covering counts, sums, averages, and distinct counts	Done

6. Final Dashboard — Screenshot

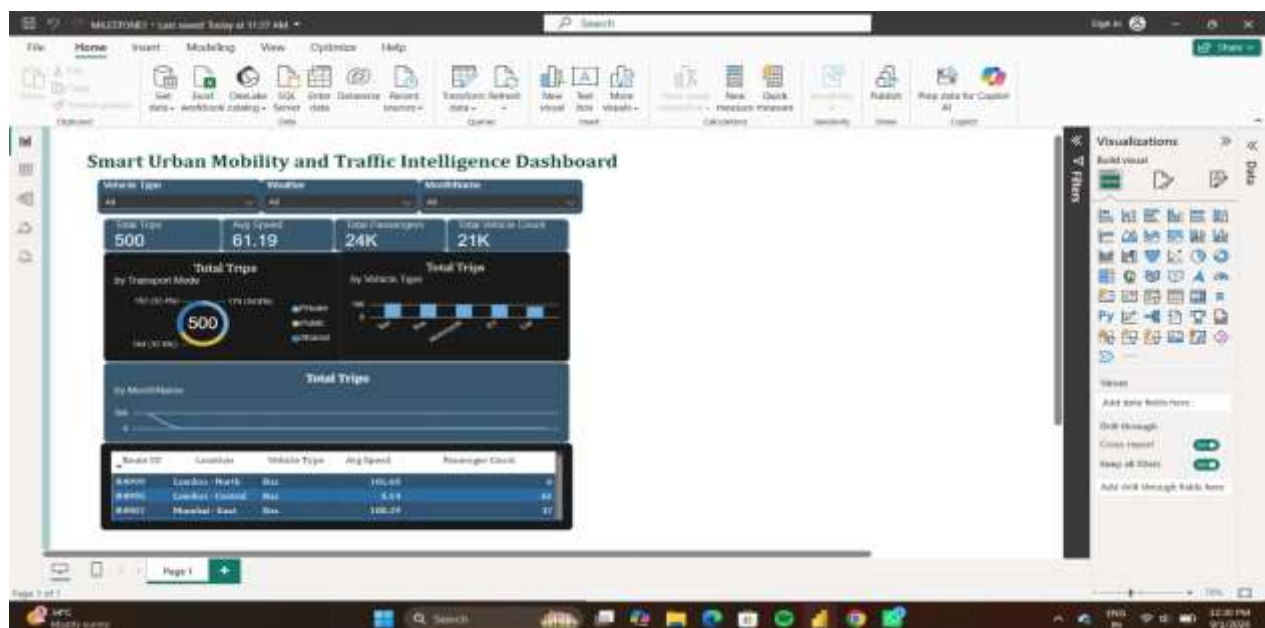


Figure 1: Milestone 1 dashboard page in Power BI Desk

Milestone 2 — Geospatial & Transit Analytics

1. Objective

Milestone 2 builds on the data model created in Milestone 1, extending it with geospatial capability and transit performance analysis. The guide's instructions broke this down into four focus areas — Geospatial Analysis, Demand Analysis, Route Performance, and Drill-down Analysis — delivered through five specific visual requirements:

- Demand Heatmap — Map visual showing trip demand concentration by city
- Mobility Flow Map — Map visual showing mobility pattern distribution
- Transit Performance — Cards (Total Trips / Passengers / Reliability / Disruptions), a route-wise Bar Chart, and a Line Chart of performance over time
- Geographic Drill-down — Map visual with a City → Region drill hierarchy
- Weather and Time Analysis — Slicers for Date, Route, Location, and Weather

2. Data Preparation for Geospatial Analysis

The existing dataset had no coordinate fields, so this phase focused on making the data map-ready before any visuals could be built.

2.1 Splitting Location into City and Region

- Re-split the combined 'Location' column (e.g., 'Tokyo - West') in Dim_location into separate 'City' and 'Region' columns using Power Query's Split Column by Delimiter.
- Applied Trim (Transform → Format → Trim) to both columns to remove hidden leading/trailing spaces left over from the split — this was the root cause of an initial failed merge that returned null coordinates.

2.2 Adding Latitude and Longitude

- Created a small reference table, CityCoordinates, with real-world Latitude/Longitude values for the dataset's five cities (Tokyo, Mumbai, New York, London, Singapore), entered via Power Query's 'Enter Data'.
- Merged CityCoordinates into Dim_location using a Left Outer join on the City column, then expanded the Latitude and Longitude fields into the table.
- Considered an alternative using the original Kaggle source file (which includes true per-route Latitude/Longitude), matched on Route ID and merged into Dim_route for higher geographic precision — documented here as a possible future refinement.

2.3 Data Categories and Summarization

- Set the Data Category of 'City' to City, 'Latitude' to Latitude, and 'Longitude' to Longitude
(**Table view → Column tools → Properties**) so Power BI's Map visual correctly interprets these fields.
- Set the Default Summarization of Latitude and Longitude to 'Do not summarize', preventing Power BI from incorrectly summing coordinate values in visuals.

2.4 Enabling Map Visuals

- Power BI Desktop disables Map and Filled Map visuals by default for security reasons. Enabled them via File → Options and settings → Options → Global → Security, and restarted Power BI Desktop for the change to take effect.

3. New DAX Measures

Milestone 2 introduces two new measures to support the Transit Performance visuals, since 'Reliability' and 'Disruptions' are not present as raw columns in the source data:

Measure	DAX Formula	Purpose
Disruptions	<code>CALCULATE(COUNTROWS(Fact_mobility), KEEPFILTERS(Fact_mobility[Weather] IN { "Storm", "Rain", "Fog" }))</code>	Counts trips occurring during disruptive weather (Storm/Rain/Fog) — a proxy metric, since the dataset has no direct disruption flag.
Reliability %	<code>DIVIDE([Total Trips] - [Disruptions], [Total Trips], 0)</code>	Percentage of trips that occurred without disruptive weather conditions.

Debugging note: the Disruptions measure initially overrode the Weather slicer's filter instead of respecting it, because an explicit CALCULATE filter on a column replaces any existing filter on that same column by default. When the Weather slicer was set to a single value (e.g., 'Fog'), this caused Disruptions to be counted across a broader set of rows than Total Trips, producing an impossible negative Reliability % (-193.75%). Adding KEEPFILTERS to the CALCULATE condition fixed this by making the measure intersect with the slicer's selection rather than replace it.

4. Dashboard Visuals Built

4.1 Demand Heatmap

- Map visual using Dim_location's Latitude/Longitude fields.
- Total Trips measure drives Bubble size, so cities with more recorded trips show larger bubbles.
- City drives the Legend, giving each city a distinct color and a legend key.

4.2 Mobility Flow Map

- The dataset contains only a single Location per trip, with no separate origin and destination fields. Drawing literal flow arcs would have required fabricating destination points, which would misrepresent the data.
- Instead, this was built as a second Map visual with the same Latitude/Longitude and Total Trips bubble size, but with Transport Mode (Public / Shared / Private) driving the Legend — shown as pie-chart bubbles per city. This honestly represents how transport modes are distributed geographically, in place of a literal, unsupported flow map.

4.3 Transit Performance

- 4 KPI Cards: Total Trips, Total Passengers, Reliability %, Disruptions.
- Bar Chart: Avg Speed by City, showing comparative route/city-level performance.
- Line Chart: Avg Speed and Reliability % by Month (MonthName, sorted by MonthNum), showing performance trends over time.

4.4 Geographic Drill-down

- A third Map visual with City and Region both placed in the Location field well, forming a City → Region drill hierarchy.
- Clicking the drill-down icon and a city bubble expands that city into its individual regions (e.g., Tokyo splits into Tokyo - West, Tokyo - East, etc.), with Total Trips sizing each bubble at every level.

4.5 Weather and Time Analysis (Slicers)

- Date slicer (from DateTable) for filtering by date range.
- Route ID slicer (from Dim_route).
- City / Location slicer (from Dim_location).
- Weather slicer (from Fact_mobility).
- All slicers were tested to confirm they correctly filter every card, chart, and map on the page, validating that the underlying star-schema relationships from Milestone 1 are functioning correctly.

5. Visual Design Enhancements

To make the dashboard more presentable and closer to a production-quality analytics product, the following formatting improvements were applied:

- Applied a consistent Power BI theme across all visuals for unified coloring.
- Styled KPI cards with a shared background color and a colored accent border, replacing plain default boxes.
- Added subtle shadows to every visual (Format → Effects → Shadow) for visual depth.
- Set a solid dark navy canvas background for a modern, cohesive look.
- Added a banner-style header behind the page title using a filled rectangle shape.

6. Requirement Checklist

Requirement	What Was Done	Status
Demand Heatmap	Map visual: Latitude/Longitude, Bubble size = Total Trips, Legend = City	Done
Mobility Flow Map	Map visual: Bubble size = Total Trips, Legend = Transport Mode (pie bubbles per city)	Done
Transit Performance — Cards	4 cards: Total Trips, Total Passengers, Reliability %, Disruptions	Done

Transit Performance — Bar Chart	Avg Speed by City	Done
Transit Performance — Line Chart	Avg Speed and Reliability % by Month	Done
Geographic Drill-down	Map with City → Region drill hierarchy	Done
Date Slicer	From DateTable	Done
Route Slicer	From Dim_route (Route ID)	Done
Location Slicer	From Dim_location (City)	Done
Weather Slicer	From Fact_mobility	Done

7. Final Dashboard — Screenshot

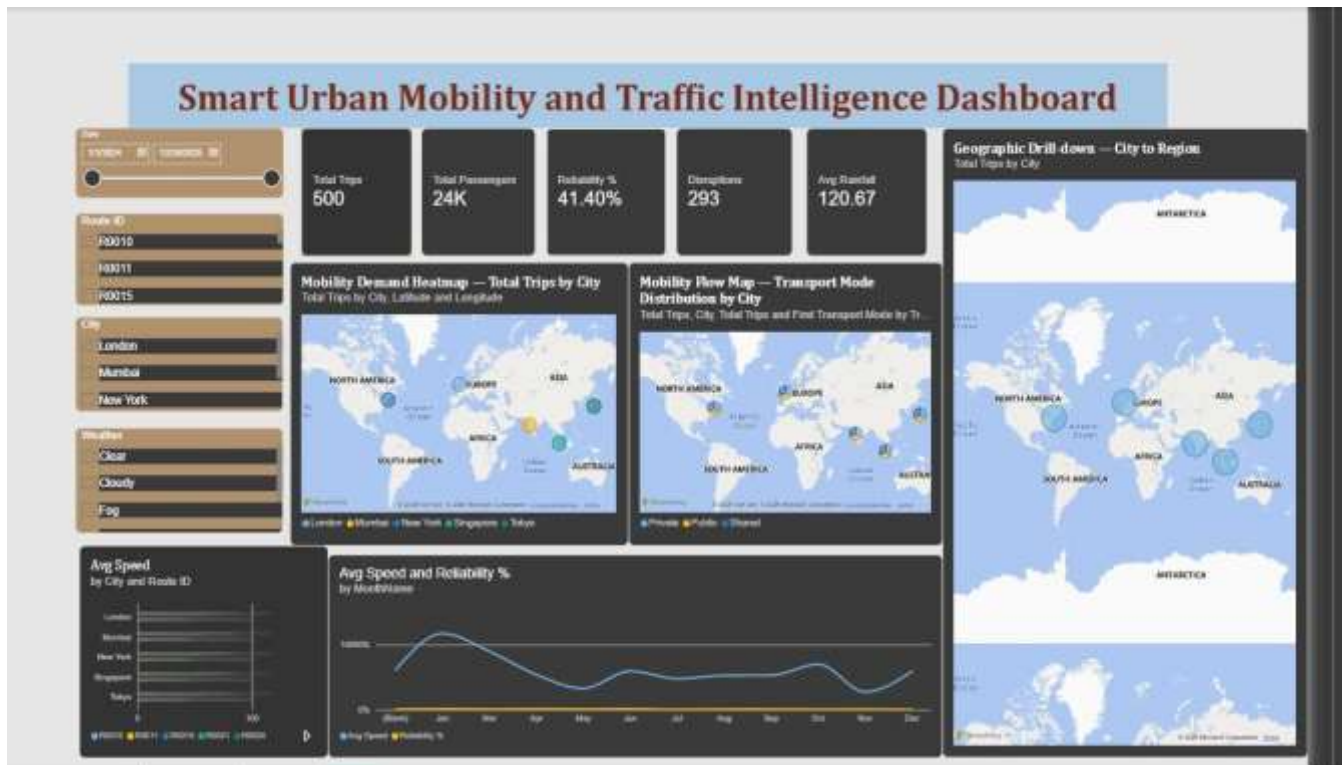


Figure 2: Milestone 2 geospatial and transit performance dashboard in Power BI Desktop

Milestone 3 — Demand Intelligence & Mobility Analytics

1. Milestone Scope

Milestone 3, as defined in the project plan, covers five deliverables:

- Build ride-hailing intelligence dashboards.
- Develop cycling and micromobility analytics.
- Implement modal substitution analysis.
- Add weather sensitivity metrics.
- Add demand forecasting metrics.

This document records the work completed across all five pages of the report: the data model, the DAX measures created, the visuals built on each page, issues diagnosed and resolved along the way, and the layout standard applied throughout. With Page 5 complete, Milestone 3 is fully delivered.

2. Data Model

The report is built on a star schema:

- Fact_mobility — one row per vehicle trip record (fact table).
- Dim_route — route reference dimension, joined to Fact_mobility on Route ID.

Key fields on Fact_mobility used across Milestone 3:

- Date, Route ID, Location, Mobility Segment (calculated column)
- Average Speed (KMH), Passenger Count, Rainfall (MM), Temperature (C)
- Weather, Weather Group, Day of Week, Day Number
- Reliability %, Disruptions, Population, Population Density, Distinct Locations, Distinct Routes

Note: an earlier draft of this documentation referenced a Vehicle Count field. That field does not exist in the live Fact_mobility table and was dropped from all measures in favour of fields confirmed present in the model (see Section 4).

2.1 Mobility Segment — Segmentation Logic

A calculated column classifies every trip into one of four segments, based on Vehicle Type and Transport Mode:

```
Mobility Segment =  
SWITCH(  
    TRUE(),  
    Fact_mobility[Vehicle Type] = "Bus"  
        || Fact_mobility[Transport Mode] = "Public", "Mass Transit",  
    Fact_mobility[Vehicle Type] IN {"Motorcycle", "EV"}, "Micromobility",  
    Fact_mobility[Vehicle Type] IN {"Taxi", "Car"}, "Ride-Hailing",  
    "Other")
```

This single column drives the page-level filters on Pages 1 and 2, and is used directly as a legend / axis field on Page 3.

3. Page 1 — Ride-Hailing Intelligence

3.1 Page Filter

Field / Setting	Value
Filter field	Mobility Segment
Filter level	Filters on this page
Selected value	Ride-Hailing
Excluded values	Mass Transit, Micromobility, Other

3.2 DAX Measures

Because the page filter already restricts the data to Ride-Hailing, each measure stays simple — the filter context handles the segmentation:

```
Ride-Hailing Trips = COUNTROWS(Fact_mobility)
Avg Speed = AVERAGE(Fact_mobility[Average Speed (KMH)])
Avg Passenger Count = AVERAGE(Fact_mobility[Passenger Count])
```

3.3 Visuals Built

Visual	Detail
KPI Card — Total Trips	Ride-Hailing Trips → 500
KPI Card — Avg Speed	Avg Speed → 61.19 KMH
KPI Card — Avg Passengers	Avg Passenger Count → 48.45
Line chart	Ride-Hailing Trips by Month
Bar chart	Ride-Hailing Trips by City (Location), sorted descending
Column chart	Ride-Hailing Trips by Weather
Table	Route ID, Ride-Hailing Trips, Avg Speed, Avg Passenger Count (with Total row)

3.4 Issue Diagnosed & Resolved — Blank KPI Card

Symptom: the first KPI card showed no value, even though the correct measure was placed in the Values field.

Diagnostic steps taken, in order:

- Confirmed the underlying data loads correctly by dragging a raw column into a Table visual — it displayed rows normally, ruling out a data or relationship problem.
- Created a zero-filter test measure (COUNTROWS with no conditions) and placed it on a new card — it was also blank, ruling out an error in the Ride-Hailing-specific DAX logic.
- Checked Filters on this visual for a stray self-filter, and confirmed report and page filters were not suppressing all rows.
- Concluded the issue was isolated to the visual instance itself rather than the data model or formulas.

Resolution: the blank card was deleted and a fresh Card visual was created and resized; the new instance rendered correctly. The same fix was applied to all three production KPI cards.

3.5 Issue Diagnosed & Resolved — Line Chart Axis Mismatch

Symptom: the "Ride-Hailing Trips by Year" line chart showed axis values (12–14) inconsistent with the 500-trip KPI total.

Cause: Power BI had defaulted the axis to an implicit Count of Date rather than the Ride-Hailing Trips measure, and the field was aggregated at Year level with the auto-generated date hierarchy rather than Month.

Resolution: rebuilt the chart at Month level — Date dragged to the X-axis with the hierarchy expanded to Month, and the Ride-Hailing Trips measure explicitly dragged into Values rather than relying on auto-summarisation of a raw column.

3.6 Layout & Formatting

- Enabled View → Gridlines and Snap to Grid for consistent alignment.
- Aligned and evenly distributed the three KPI cards across the top of the page.
- Aligned and evenly distributed the Month / City / Weather chart row beneath the KPI cards.
- Widened the table visual and its column borders so the route-level table fills the available width with no blank space or unnecessary scrolling.

4. Page 2 — Cycling & Micromobility Analytics

Built as a fresh page mirroring Page 1's structure, rather than a duplicate of Page 1 — this avoided the stale-title problem duplication causes (visual titles and field labels carry over as static text and do not update to reflect a changed filter).

4.1 Page Filter

Field / Setting	Value
Filter field	Mobility Segment
Filter level	Filters on this page

Field / Setting	Value
Selected value	Micromobility
Excluded values	Mass Transit, Ride-Hailing, Other

4.2 DAX Measures

Measures were named generically ("Segment __" rather than "Ride-Hailing __") so the same three measures can be reused on Page 3 without renaming — only the page filter changes.

```
Segment Trips = COUNTROWS(Fact_mobility)
Segment Avg Speed = AVERAGE(Fact_mobility[Average Speed (KMH)])
Segment Avg Passenger Count = AVERAGE(Fact_mobility[Passenger Count])
```

Note: a Vehicle Count measure was initially planned for the third KPI, but no Vehicle Count field exists on Fact_mobility. Passenger Count was used instead to keep Page 2 structurally identical to Page 1; Rainfall/Temperature correlation was deferred to Page 4, where it belongs per the milestone scope.

4.3 Visuals Built

Visual	Detail
KPI Card — Total Trips	Segment Trips → 137
KPI Card — Avg Speed	Segment Avg Speed → 64.65 KMH
KPI Card — Avg Passengers	Segment Avg Passenger Count → 49.26
Line chart	Micromobility Trips by Month
Bar chart	Micromobility Trips by City (Location), sorted descending
Column chart	Micromobility Trips by Weather
Table	Route ID, Segment Trips, Segment Avg Speed, Segment Avg Passenger Count (with Total row)

4.4 Verification

Segment split confirmed via the page filter pane: Mass Transit 238, Micromobility 137, Ride-Hailing 125 — the Micromobility KPI card total (137) matches exactly, confirming the filter and measures are wired correctly

5. Page 3 — Modal Substitution Analysis

Unlike Pages 1–2, this page carries no page-level Mobility Segment filter. The purpose is to compare segments against each other — as a share of total trips — rather than isolate one segment, so Mobility Segment is used as a dimension (legend / axis) rather than a filter.

5.1 DAX Measures

```
Total Trips (All Segments) =  
CALCULATE(  
    COUNTROWS(Fact_mobility),  
    ALL(Fact_mobility[Mobility Segment])  
)
```

```
Segment Share % =  
DIVIDE(  
    COUNTROWS(Fact_mobility),  
    [Total Trips (All Segments)]  
)
```

Total Trips (All Segments) uses ALL() to ignore whatever segment filter context a visual applies, always returning the grand total; Segment Share % then divides the in-context segment count by that constant total, giving each segment's correct share inside any visual — a legend, an axis, or a visual-level filter.

5.2 KPI Cards

Three cards, each carrying its own visual-level filter on Mobility Segment (not a page-level filter, so each card can show a different segment slice independently):

Card	Measure	Visual-level filter
Micromobility Trips	Segment Trips	Mobility Segment = Micromobility
Ride-Hailing Trips	Segment Trips	Mobility Segment = Ride-Hailing
Micromobility Share %	Segment Share % (formatted as Percentage)	Mobility Segment = Micromobility

5.3 Segment Share by Weather

100% stacked column chart — Weather on the X-axis, Mobility Segment as Legend, Segment Trips in Values. This is the core substitution visual: it surfaces whether Micromobility's share shrinks and Ride-Hailing's share grows under Rain/Storm conditions relative to Clear.

5.4 Segment Share Trend by Month

100% stacked area chart — Date (Month) on the X-axis, Mobility Segment as Legend, Segment Trips in Values. Shows how the substitution pattern moves over the course of the year rather than at a single point in time.

5.5 Segment Comparison Table

Table with columns: Location, Mobility Segment, Segment Trips, Segment Share %; Total row enabled. Because the table has both Location and Mobility Segment in row context, Segment Share % resolves to

each segment's share within that city (rows sum to ~100% per city) rather than share of the full dataset — confirmed as the intended behaviour.

6. Page 4 — Weather Sensitivity

Covers trip volume and speed across Weather / Weather Group, plus a rainfall correlation view, using the Rainfall (MM) and Temperature (C) fields already present on Fact_mobility. No page-level segment filter — this page looks at weather effects across all trips, or per segment via the existing Segment measures.

6.1 DAX Measures

```
Avg Rainfall (MM) = AVERAGE(Fact_mobility[Rainfall (MM)])  
Avg Temperature (C) = AVERAGE(Fact_mobility[Temperature (C)])
```

Segment Trips and Segment Avg Speed (from Page 2) are reused unchanged.

6.2 KPI Cards

Card	Measure
Avg Rainfall	Avg Rainfall (MM)
Avg Temperature	Avg Temperature (C)
Avg Speed (All Trips)	Segment Avg Speed, no visual filter

6.3 Visuals Built

- Clustered column chart — "Trip Volume by Weather Group": Weather Group on X-axis, Segment Trips in Values.
- Line-and-clustered-column combo — "Speed vs Trip Volume by Weather": Weather on X-axis, Segment Trips as columns, Segment Avg Speed as a secondary-axis line — shows whether speed drops as volume rises in adverse weather (congestion) or moves independently.
- Scatter chart — "Speed vs Rainfall Correlation": Avg Rainfall (MM) on X-axis, Segment Avg Speed on Y-axis, Date (Month) in Details, one point per month.
- Table — "Weather Group summary": Weather Group, Segment Trips, Avg Temperature (C), Segment Avg Speed, Total row enabled.

6.4 Layout

- Gridlines and Snap to Grid enabled.
- KPI row aligned and distributed across the top.
- Volume and speed-sensitivity charts placed side by side beneath the KPI row.
- Scatter chart and summary table fill the bottom row.

7. Page 5 — Demand Forecasting

The final page of the milestone, covering trend-based forecasting and day-of-week seasonality. No page-level segment filter — the base trend uses combined trip volume, with segment-level slicing available as an optional add-on rather than the default view.

7.1 Base Trend Visual

Line chart — Date (Month) on the X-axis, Segment Trips (or a new unfiltered Total Trips = COUNTROWS(Fact_mobility) measure) in Values. Titled "Demand Trend".

7.2 Built-in Forecast

Applied via the Analytics pane on the line chart (Analytics → Forecast → Add):

- Forecast length: 3–6 periods forward (months)
- Confidence interval: 95% (default)
- Seasonality: set explicitly (e.g. 12 for a yearly monthly cycle) rather than left on Auto, since Auto can miss weekly patterns in trip data

This produces a dotted projection line with a shaded confidence band extending past the last actual data point.

7.3 Day-of-Week Seasonality

Column chart — Day of Week on the X-axis, Segment Trips in Values. Titled "Demand by Day of Week".

Issue diagnosed & resolved: Power BI's default alphabetical sort on Day of Week (Friday, Monday, Saturday...) misrepresented the weekly cycle. Fixed via right-click on the Day of Week field in the Data pane → Sort by column → Day Number, restoring correct Monday–Sunday order.

7.4 KPI Cards

Two cards used rather than three, to avoid forcing an unclear third metric onto the page: "Total Trips to Date" (Segment Trips / Total Trips) and "Avg Speed (All Trips)" (Avg Speed, reused from Page 1).

7.5 Layout

- Gridlines and Snap to Grid enabled.
- KPI row across the top.
- Demand Trend (with forecast) as the larger visual beneath the KPI row.
- Day-of-Week seasonality chart placed alongside or below the trend visual.

8. Milestone 3 Status Summary

Milestone 3 sub-task	Status	Delivered on
Build ride-hailing intelligence dashboards	Complete	Page 1
Develop cycling and micromobility analytics	Complete	Page 2
Implement modal substitution analysis	Complete	Page 3
Add weather sensitivity metrics	Complete	Page 4
Add demand forecasting metrics	Complete	Page 5

All five Milestone 3 deliverables are complete.

9. Final Dashboard Screenshots

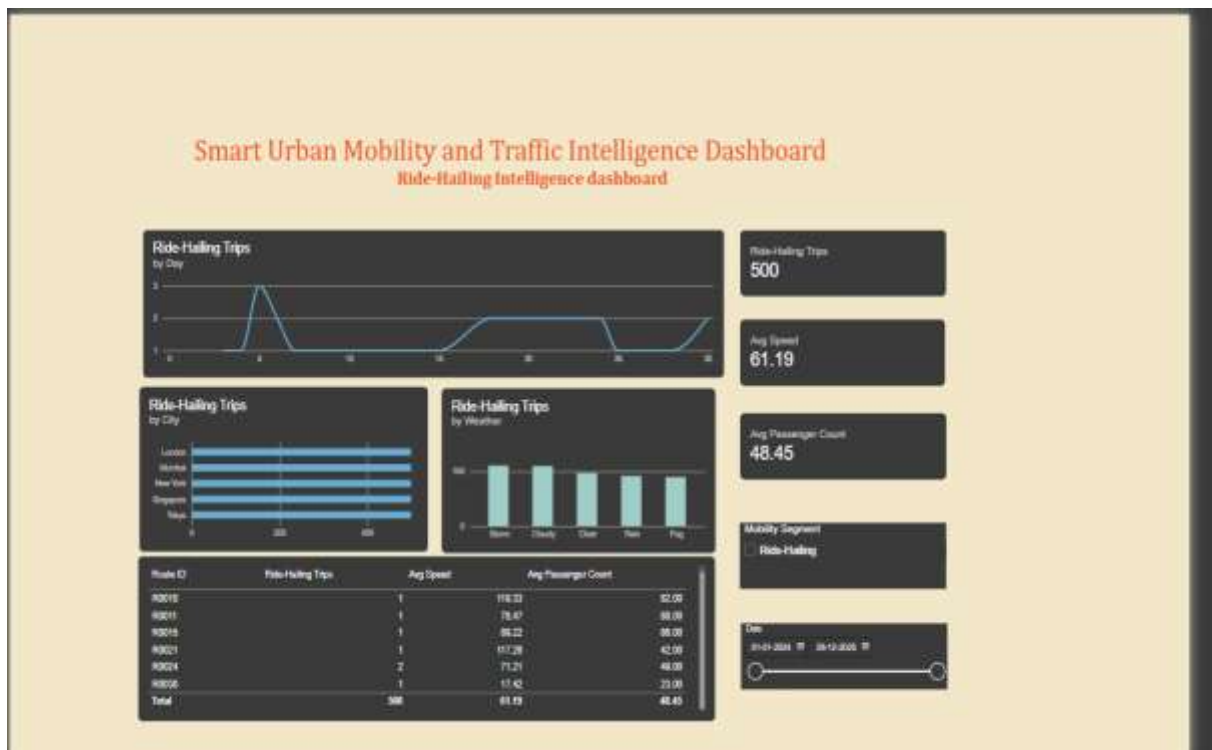


Figure 3: Ride-Hailing Intelligence Dashboard in PowerBI



Figure 4: Cycling & Micromobility Dashboard in PowerBI

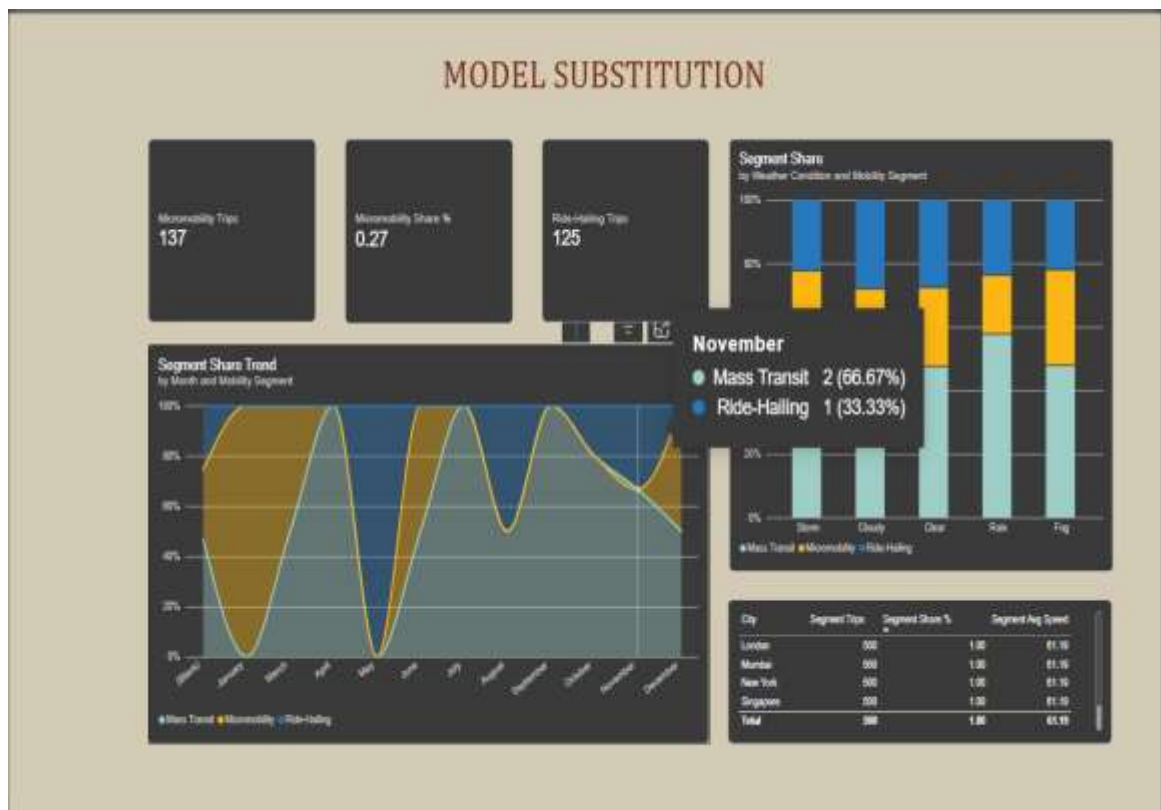


Figure 5: Model Substitution Dashboard in PowerBI

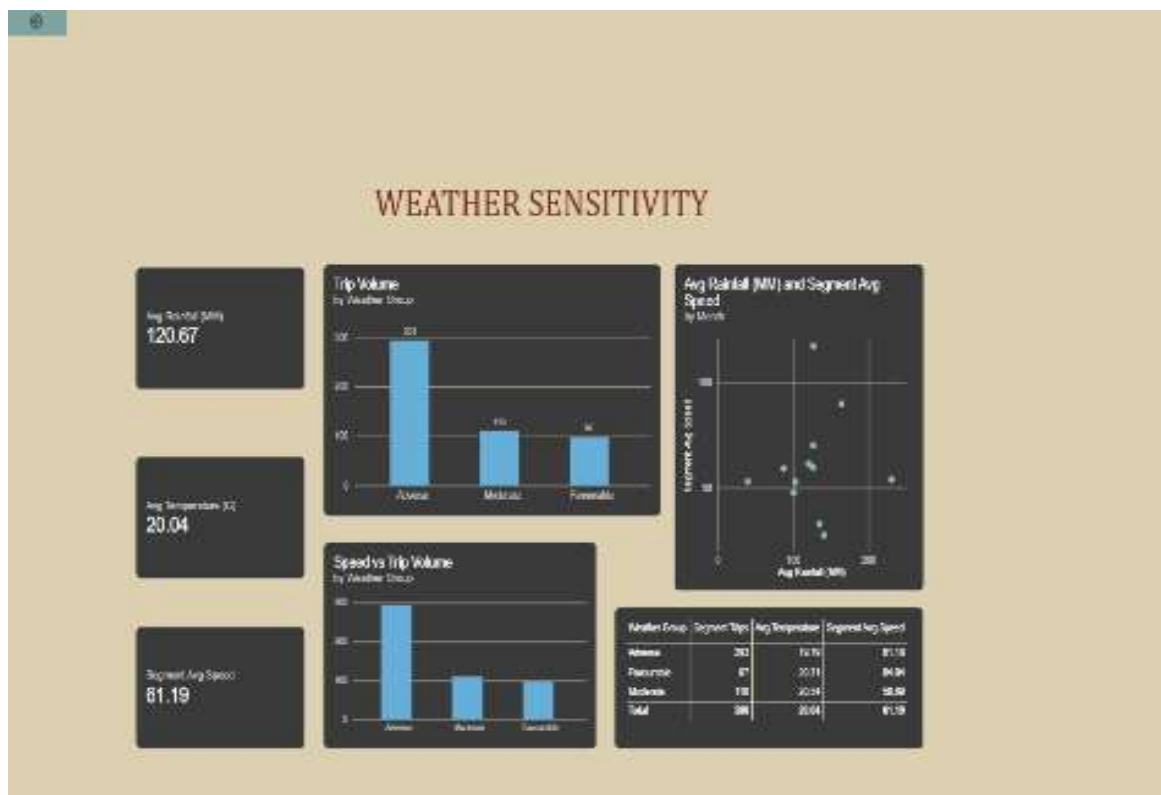


Figure 6: Weather Sensitivity Dashboard in PowerBI

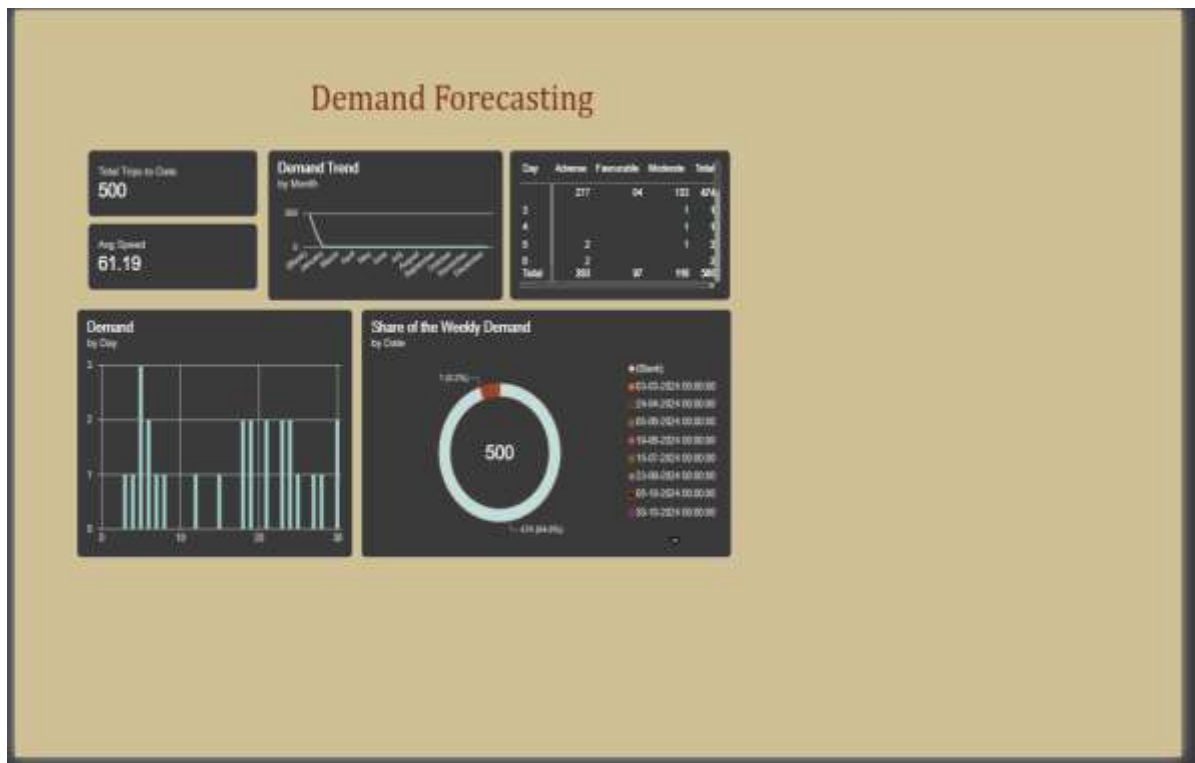


Figure 7: Demand Forecasting Dashboard in PowerBI

Milestone 4 — Equity, Forecasting & Finalization

1. Milestone Scope

Milestone 4, as defined in the project plan, covers four deliverables:

- Implement Mobility Access Index and equity dashboards.
- Build executive intelligence dashboards.
- Add forecasting, anomaly detection, and reporting features.
- Optimize dashboard performance and complete deployment readiness.

This document records the work completed toward the first three deliverables — the Mobility Access & Equity page, the Executive Intelligence page, and the Forecasting & Anomalies page — including new measures created, model issues diagnosed and resolved along the way, and the visuals built on each page. Reporting features, performance optimization, and deployment readiness remain open and are covered under Next Steps in Section 6.

2. Data Model Issues Diagnosed & Resolved

Two model-level issues surfaced while building this milestone. Both were resolved at the source rather than worked around in individual visuals, since they would otherwise have affected every page built afterward.

2.1 Population* / Population Density* are not per-location values

Symptom: a planned Trips Per Capita measure produced almost no variation across locations.

Diagnosis: a table with Min, Max, and Average of Population* by Location showed Min and Max far apart within every single location (e.g. Min 6.92 / Max 93.29 for one location) — confirming Population* is a per-trip value, not a fixed population figure per location.

Resolution: the Trips Per Capita component was dropped from the Mobility Access Index. Trip volume (Segment Trips) was used in its place as the access/service-level component.

2.2 Fact_mobility[Date] stored as Date/Time, not Date

Symptom: the Demand Trend line chart showed a large (Blank) category, a short real segment under January, and a flat line at 0 for every other month.

Diagnosis: Fact_mobility's Date column held a full date and time (e.g. 17-11-2024 14:00:00), while the separate DateTable stored plain dates at midnight. The relationship between the two could only match rows that happened to fall exactly at midnight, so nearly all rows failed to join.

Resolution: in Power Query, Fact_mobility's Date column was changed to type Date (not Date/Time), truncating the time portion so it joins cleanly to DateTable.

2.3 Duplicate Route ID blocking model reload

Symptom: after applying the date-type fix, the model failed to reload with cascading errors on Fact_mobility, Dim_location, and Dim_route.

Diagnosis: Dim_route contained two rows with the same Route ID ("R1070"). Dim_route sits on the one side of a one-to-many relationship with Fact_mobility, which requires unique values on that side; the duplicate blocked the entire model refresh.

Resolution: the duplicate Route ID was located by sorting the Route ID column in Power Query and removed via Remove Duplicates, after confirming the two rows were identical. The model then reloaded successfully and the (Blank) category disappeared from downstream visuals.

2.4 Sparse daily grain vs. Month-level analysis

Symptom: even after both fixes, the Demand Trend and Demand Anomalies charts still showed a near-flat line at 1–2.

Diagnosis: a check of trip counts per individual date showed almost every date carrying exactly 1 trip (500 trips spread across ~500 distinct dates) — too sparse for a meaningful daily trend or anomaly signal.

Resolution: both line charts were rebuilt on DateTable[Date] drilled to Month level, while keeping the axis type Continuous (required for Forecast and Anomalies), rather than plotting raw daily dates.

3. Page 8 — Mobility Access Index & Equity

3.1 Component Measures

```
Avg Reliability % = AVERAGE(Fact_mobility[Reliability %])
Route Density =
DIVIDE(
    DISTINCTCOUNT(Fact_mobility[Route ID]),
    AVERAGE(Fact_mobility[Population Density*])
)
```

3.2 Composite MAI Score

All three components are normalized to a 0–1 scale relative to the lowest and highest location, then averaged equally — trip volume replacing the originally planned Trips Per Capita per Section 2.1:

```
MAI Score =
VAR TripsNorm =
    DIVIDE(
        [Segment Trips] - MINX(ALL(Fact_mobility[Location]), [Segment Trips]),
        MAXX(ALL(Fact_mobility[Location]), [Segment Trips])
        - MINX(ALL(Fact_mobility[Location]), [Segment Trips])
    )
VAR ReliabilityNorm =
    DIVIDE(
        [Avg Reliability %] - MINX(ALL(Fact_mobility[Location]), [Avg Reliability %]),
        MAXX(ALL(Fact_mobility[Location]), [Avg Reliability %])
        - MINX(ALL(Fact_mobility[Location]), [Avg Reliability %])
    )
VAR RouteNorm =
    DIVIDE(
        [Route Density] - MINX(ALL(Fact_mobility[Location]), [Route Density]),
        MAXX(ALL(Fact_mobility[Location]), [Route Density])
        - MINX(ALL(Fact_mobility[Location]), [Route Density])
    )
RETURN
    (TripsNorm + ReliabilityNorm + RouteNorm) / 3
```

3.3 Visuals Built

Visual	Detail
KPI Card	Overall MAI Score
KPI Card	Avg Reliability
KPI Card	Avg Speed
KPI Card	Total Trips
Bar chart	MAI Score by Location, sorted ascending (lowest-access first), conditional color scale
Scatter chart	Population Density* (avg) vs Segment Trips, bubble size = Population* (avg), detail = Location
Table	Location, Population* (avg), Segment Trips, Avg Reliability %, MAI Score with

Visual	Detail
	conditional formatting
Slicers	Mobility Segment, Weather Group

Locations in the dataset are city-zone combinations (e.g. London – Central, Mumbai – East) spanning roughly 20 distinct locations across four to five cities.

4. Page 9 — Executive Intelligence Dashboard

A rollup page — smaller versions of the strongest visuals from prior pages, built for a viewer who will not open every individual page. No page-level filter, so all segments and locations are visible together.

4.1 Supporting Measure

All Trips Count = COUNTROWS(Fact_mobility)

Unlike Total Trips (All), which ignores segment context via ALL(), this measure respects whatever segment context a visual applies — needed so the segment-share donut can split correctly.

4.2 Visuals Built

Visual	Detail
KPI Card	Total Trips (Total Trips (All))
KPI Card	Avg Speed
KPI Card	Avg Reliability
KPI Card	Overall MAI Score
KPI Card	Avg Rainfall
Donut chart	Trips by Segment (Mobility Segment / All Trips Count)
Line chart	Demand Trend (Date by Month / All Trips Count)
Column chart	Demand by Weather (Weather / All Trips Count)
Bar chart	MAI Score by Location — condensed summary version of Page 8's chart

5. Page 10 — Forecasting & Anomaly Detection

5.1 Anomaly Measures

```
Speed Anomaly Flag =
VAR AvgSpd = [Segment Avg Speed]
VAR OverallAvg = CALCULATE([Segment Avg Speed], ALL(Fact_mobility))
VAR OverallStdDev = CALCULATE(STDEV.P(Fact_mobility[Average Speed (KMH)]), ALL(Fact_mobility))
RETURN
    IF(ABS(AvgSpd - OverallAvg) > (2 * OverallStdDev), "Anomaly", "Normal")
```

```
Anomaly Count =
CALCULATE(
    COUNTROWS(Fact_mobility),
```

```
FILTER(Fact_mobility, [Speed Anomaly Flag] = "Anomaly")
)
```

Note: Anomaly Count initially failed with "a function 'PLACEHOLDER' has been used in a True/False expression that is used as a table filter expression" when Speed Anomaly Flag — a measure, not a column — was passed directly as a CALCULATE filter argument. Because a measure must be evaluated row by row, the fix was to wrap the comparison in FILTER(Fact_mobility, [Speed Anomaly Flag] = "Anomaly") rather than using it as a bare boolean filter.

5.2 Demand Trend (Forecast)

- Line chart — DateTable[Date] drilled to Month, axis type set to Continuous, single measure (All Trips Count) in Values, no legend.
- Built-in Forecast added via the Analytics pane: forecast length 3–6 periods, 95% confidence interval, seasonality set explicitly rather than left on Auto.

5.3 Demand Anomalies

- A second, separate line chart with the same axis and Values configuration as the trend chart — Forecast and Anomalies are not combined on one visual.
- Anomaly detection enabled via the Analytics pane at Month-level granularity, with Sensitivity adjustable (started at 70).
- Explanatory fields (Weather, Mobility Segment, Location) available via "Explain this anomaly" on any flagged point.

5.4 Visuals Built

Visual	Detail
KPI Card	Total Trips (All Trips Count)
KPI Card	Anomalies Flagged (Anomaly Count)
KPI Card	Avg Rainfall
Line chart	Demand Trend, with built-in Forecast
Line chart	Demand Anomalies, with built-in anomaly detection

6. Milestone 4 Status Summary

Milestone 4 sub-task	Status	Delivered on
Implement Mobility Access Index and equity dashboards	Complete	Page 8
Build executive intelligence dashboards	Complete	Page 9
Add forecasting and anomaly detection features	Complete	Page 10

7. Open Items & Next Steps

- Add page navigation buttons (Insert → Buttons → Navigator → Page navigator) consistently across all ten pages.
- Enable synced slicers (View → Sync slicers) for Mobility Segment, Location, and Year across relevant pages.
- Build an Area/Location drill-through page for per-location detail.
- Run Performance Analyzer, address any slow visuals, confirm single-direction relationships, and cap visuals per page.
- Apply a consistent theme across all pages, set up Row-Level Security if required, configure scheduled refresh, and publish to the workspace.

8. Final Dashboards



Figure 8: Mobility Access Index Dashboard in PowerBI



Figure 9: Executive Intelligence Dashboard in PowerBI

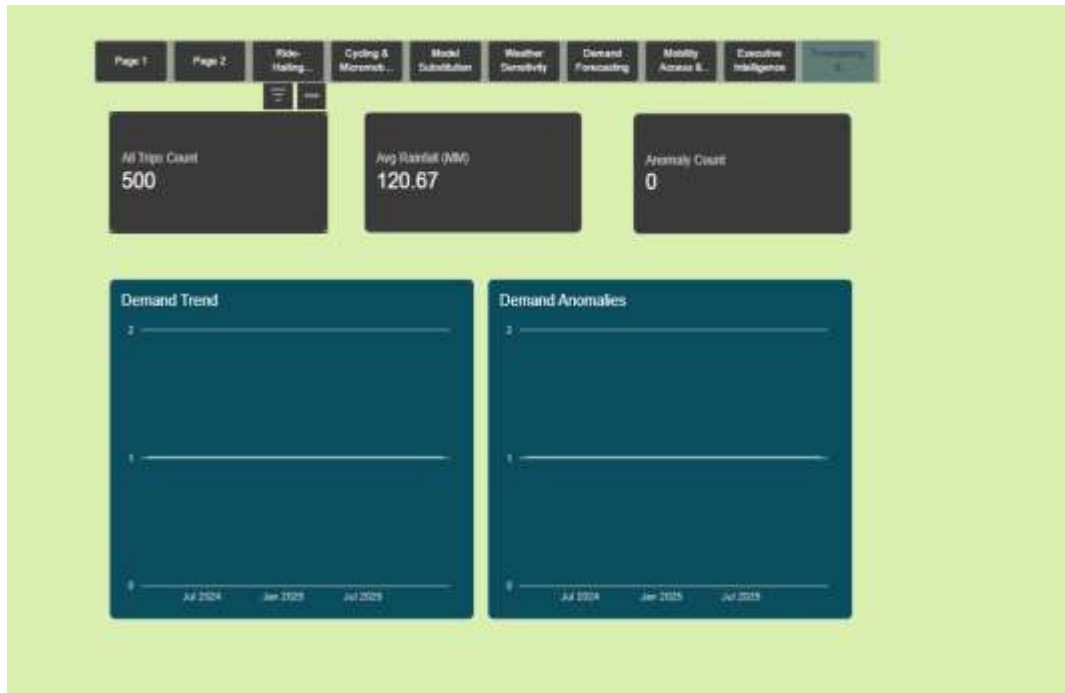


Figure 10: Forecasting Dashboard in PowerBI

➤ Conclusion

- The **Smart Urban Mobility and Traffic Intelligence Dashboard** project was developed in Power BI to analyze urban mobility data and provide meaningful insights into transportation patterns.
- The project was completed progressively through four milestones.
- In **Milestone 1**, I prepared the dataset, created the star-schema data model, established relationships, developed DAX measures, and built the initial dashboard with KPIs, charts, slicers, and detailed tables.
- In **Milestone 2**, I extended the dashboard with geospatial and transit analysis. I added demand heatmaps, mobility distribution maps, geographic drill-down, transit performance analysis, and weather and time-based filtering.
- In **Milestone 3**, the project was expanded into demand intelligence and mobility analytics. I developed separate analysis for ride-hailing, micromobility, modal substitution, weather sensitivity, and demand forecasting.
- Finally, **Milestone 4** focused on mobility access and equity, executive-level reporting, forecasting, and anomaly detection. The Mobility Access Index was created using trip volume, reliability, and route density, while the Executive Intelligence Dashboard provided a consolidated view of the major mobility indicators. Forecasting and anomaly detection were also added to support analysis of future demand and unusual patterns.

Throughout the project, several data and modelling issues were identified and resolved, including date-type mismatches, duplicate route IDs, population-data limitations, and sparse daily data. These corrections helped maintain reliable relationships and improve the accuracy of the dashboard analysis.

- Overall, the project provided practical experience in **data preparation, data modelling, DAX, interactive visualization, geospatial analysis, mobility segmentation, equity analysis, forecasting, and anomaly detection using Power BI**. The completed dashboards provide different levels of analysis, from detailed mobility patterns to executive-level summaries.

- The remaining activities identified in the documentation are mainly finalization and deployment tasks, including page navigation, synced slicers, drill-through analysis, performance optimization, consistent theme application, scheduled refresh, and publishing the report to the workspace.

➤ References

1. **Microsoft Power BI Desktop** — Used for data preparation, data modelling, DAX calculations, dashboard development, interactive visualizations, forecasting, and anomaly detection.
2. **Smart_City_Mobility_Sample_500.xlsx** — Primary dataset used for the Smart Urban Mobility and Traffic Intelligence Dashboard. The dataset contains 500 sampled trip records from the larger Smart City Mobility dataset.
3. **Infosys Springboard Virtual Internship — Smart Urban Mobility and Traffic Intelligence Project Documentation** — Project requirements, milestone objectives, dashboard requirements, and implementation guidelines.
4. **Microsoft Power BI Documentation** — Reference for Power BI data modelling, DAX measures, visualizations, filters, forecasting, and anomaly detection.
5. **Microsoft DAX Documentation** — Reference for DAX functions and calculations used during the development of the dashboard, including CALCULATE, FILTER, ALL, DIVIDE, AVERAGE, COUNTROWS, and DISTINCTCOUNT.
6. **Power BI Desktop Analytics Features** — Reference for built-in forecasting and anomaly detection used in the Demand Forecasting and Anomaly Detection dashboard.
7. **Project-specific Power Query transformations and DAX calculations** — Developed and implemented during the internship project for data cleaning, modelling, segmentation, mobility access analysis, forecasting, and dashboard development.